



Resuscitation Bundle in Pediatric Shock Decreases Acute Kidney Injury and Improves Outcomes

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Objective To investigate the impact of an early emergency department (ED) protocol-driven resuscitation (septic shock protocol [SSP]) on the incidence of acute kidney injury (AKI).

Study design This was a retrospective pediatric cohort with clinical sepsis admitted to the pediatric intensive care unit (PICU) from the ED before (2009, PRE) and after (2010, POST) implementation of the SSP. AKI was defined by pRIFLE (pediatric version of the Risk of renal dysfunction; Injury to kidney; Failure of kidney function; Loss of kidney function, End-stage renal disease creatinine criteria).

Results A total of 202 patients (PRE, n = 98; POST, n = 104) were included (53% male, mean age 7.7 ± 5.6 years, mean Pediatric Logistic Organ Dysfunction [PELOD] 8.9 ± 12.7 , mean Pediatric Risk of Mortality score 5.3 ± 13.9). There were no differences in demographics or illness severity between the PRE and POST groups. POST was associated with decreased AKI (54% vs 29%, $P < .001$), renal-replacement therapy (4 vs 0, $P = .04$), PICU, and hospital lengths of stay (LOS) (1.9 ± 2.3 vs 4.5 ± 7.6 , $P < .01$; 6.3 ± 5.1 vs 15.3 ± 16.9 , $P < .001$, respectively), and mortality (10% vs 3%, $P = .037$). The SSP was independently associated with decreased AKI when we controlled for age, sex, and PELOD (OR 0.27, CI 0.13-0.56). In multivariate analyses, the SSP was independently associated with shorter PICU and hospital LOS when we controlled for AKI and PELOD ($P = .02$, $P < .001$, respectively).

Conclusion A protocol-driven implementation of a resuscitation bundle in the pediatric ED decreased AKI and need for renal-replacement therapy, as well as PICU and hospital LOS and mortality. (*J Pediatr* 2015;167:1301-5).

Septic shock affects almost more than 50 000 children annually¹⁻⁴ and is the leading cause of acute kidney injury (AKI) in the pediatric intensive care unit (PICU).⁵ One recent point prevalence study in 26 nations found that 8% of children in PICUs met the criteria for severe sepsis.² Mortality rates of up to 25% have been reported in both industrialized and developing nations.² Standardized definitions and stratification schemes⁶ for AKI have greatly improved our understanding of the epidemiology and impact of AKI. Despite increased awareness and detection through the widespread use of standardized AKI definitions, AKI remains an independent mortality and morbidity risk for hospitalized patients. Twenty-five percent of patients in the PICU are discharged with AKI, and follow-up frequently is sporadic.^{7,8} Repeated episodes of AKI, especially in these with chronic conditions, are now recognized to lead to the development of chronic kidney disease (CKD).⁹⁻¹⁴

Patient outcomes have improved consistently in the PICU during the last 3 decades; however, despite advances in renal support, little has changed in the outcome of critical illness-associated AKI. Furthermore, AKI already is present on admission to the PICU in up to 30% of patients.¹⁵ Because a significant percentage of AKI already is established on admission to the PICU, we hypothesized that early intervention before admission to the PICU would reduce the incidence of AKI. Texas Children's Hospital implemented a protocol-driven sepsis resuscitation bundle (septic shock protocol [SSP]) in the emergency department (ED) in 2010, resulting in earlier recognition of shock and significant improvements in time-to-critical interventions such as fluid resuscitation with normal saline and appropriate intravenous antibiotics.¹⁶ Given the poor outcomes once AKI is established and the frequency of association with sepsis, we wanted to investigate the incidence of AKI in children receiving protocolized resuscitation in the pre-PICU setting to see whether AKI could be prevented before arrival to the PICU.

AKI	Acute kidney injury
CKD	Chronic kidney disease
ED	Emergency department
ESRD	End-stage renal disease
LOS	Length of stay
PELOD	Pediatric logistic organ dysfunction
PICU	Pediatric intensive care unit
RRT	Renal-replacement therapy
SSP	Septic shock protocol

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The authors declare no conflicts of interest.

Portions of this study were presented orally at the meeting of the European Society of Intensive Care Medicine, Paris, France, October 5-9, 2013.

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<http://dx.doi.org/10.1016/j.jpeds.2015.08.044>

Methods

We performed a retrospective review of patients with clinical septic shock admitted to the PICU from the ED in the 12 months before (2009; PRE) and the 6 months immediately after institution of the SSP (POST) in 2010. AKI was defined a priori as one of the clinical outcomes to be tracked after the implementation of the quality improvement project and was defined by pediatric version of the Risk of renal dysfunction; Injury to kidney; Failure of kidney function; Loss of kidney function, End-stage renal disease (ESRD) creatinine criteria (**Table I**).^{6,17} Severe AKI was defined as pediatric version of the Risk of renal dysfunction; Injury to kidney; Failure of kidney function; Loss of kidney function, ESRD injury or failure. Outcome criteria of loss and ESRD were not investigated. Because urine output data were not consistently available, this variable was not integrated into the AKI definition. The lowest serum creatinine available in the last 6 months was recorded for each patient to define baseline creatinine clearance. If no baseline creatinine measurement was available, the patient was assumed to have normal renal function (estimated creatinine clearance of 120 mL/min/1.73 m²).⁶ Only the initial admission for shock for each patient was analyzed. Patients with known CKD or ESRD were excluded, as were patients who had no serum creatinine obtained during the index admission. Institutional review board approval was obtained.

The details of the SSP have been published previously.¹⁶ To summarize, patients who were suspected to have septic shock or severe sepsis were triaged rapidly in the ED and received hemodynamic measure-targeted fluid resuscitation with normal saline, intravenous antibiotics, and, when applicable, vasoactive support and steroids, as part of an institutional protocol. The control cohort (PRE) consisted of patients who were admitted to the PICU from the ED with clinical septic shock in the 12 months preceding institution of shock protocol. Demographics, comorbidities, physiological and laboratory data required for calculation of severity of illness scores, the amount of fluid received in the ED, as well as outcome data (length of PICU and hospital length of stay [LOS, calculated from time of ED triage], 30-day mortality, and use of renal-replacement therapy [RRT]) were recorded.

Table I. Pediatric version of the Risk of renal dysfunction; Injury to kidney; Failure of kidney function; Loss of kidney function, ESRD (pRIFLE) classification

Category	Definition	Mortality rate (aOR)
R: Risk	Decline in eCCL >25% from baseline	4.3 (2.3-7.8)
I: Injury	Decline in eCCL >50% from baseline	8.1 (4.6-14.3)
F: Failure	Decline in eCCL >75% from baseline	15.6 (9.5-25.6)
L: Loss	Loss of renal function >4 wk	Unknown/not studied
E: ESRD	Complete loss of renal function for >3 mo	Unknown/not studied

eCCL, estimated creatinine clearance.

Pediatric modified RIFLE classification system as per Akcan-Arikan et al⁶ and reported mortality rate in terms of aORs and AKI by RIFLE.¹⁷

Continuous data were presented as mean $\pm$ SD or median (IQR). Categorical data were presented as percentages. Parametric data were analyzed using the Student *t* or χ^2 test. Nonparametric data were analyzed by the Kruskal-Wallis or Mann-Whitney *U* test. Multivariate analyses were performed with linear or logistic regression. All statistical analyses were performed using Stata 11 (Stata, Inc, College Station, Texas).

Results

Of 206 patients identified, 2 each had ESRD or lacked any serum creatinine measurements and were excluded (**Table II**). There were no differences in age, sex, presence of preexisting conditions, or Pediatric Risk of Mortality or pediatric logistic organ dysfunction (PELOD) scores between PRE and POST SSP. Overall, 83 patients (41%) had AKI (risk = 52 [63%], injury = 23 [28%], failure = 8 [10%]). There were no differences in age, sex, or the proportion of patients with comorbidities between patients with and without AKI. Patients with AKI had greater mortality, greater severity-of-illness scores, and longer PICU and hospital LOS (**Table III**). The SSP (POST) group in our study had a median time to intervention of 34 minutes. Twenty-six percent of PRE patients got the first antibiotic dose within the first 60 minutes of arrival in the ED compared with 56% of POST patients ($P < .001$). Children without AKI received more fluids than those with AKI, but the differences volume received was unlikely to be of clinical significance (56 mL/kg vs 47 mL/kg, $P = .002$).

Fewer patients developed AKI in the POST group (54% PRE vs 29% in POST, $P < .001$); there also were reductions in mortality, LOS, and use of RRT (**Table IV**). The stages of AKI were similar in PRE and POST groups. Severe AKI, defined as combination of injury and failure, was greater in the PRE than the POST group (21/98 [21%] vs 10/104 [10%], $P = .01$). POST patients received statistically more fluid than PRE patients in the ED (56 vs 49 mL/kg), but this difference is of minimal clinical significance. In multivariate analyses, the POST period was independently associated with decreased AKI when we controlled for age, sex, and PELOD or Pediatric Risk of Mortality. The shock protocol also was associated with shorter PICU and hospital

Table II. Characteristics of the study population

Variables	PRE control group (n = 98)	POST intervention group (n = 104)	P value
Age, y	8.1 $\pm$ 5.8	7.3 $\pm$ 5.5	.31
Male, n (%)	54 (55)	53 (51)	.29
Previously healthy, n (%)	38 (39)	45 (43)	.56
PRISM score	6.6 $\pm$ 14.9	4.2 $\pm$ 12.9	.22
PELOD score	7.7 $\pm$ 16.1	10.2 $\pm$ 8.4	.16
Fluid given, mL/kg	48.7 $\pm$ 24.1	55.9 $\pm$ 15.7	.01
Time to first bolus, min	65 (29-121)	34 (24-85)	.01
Time to first antibiotic, min	141 (86-199)	54 (41-125)	.001

PRISM, Pediatric Risk of Mortality.

Time to first fluid bolus and first antibiotic is time elapsed from arrival in ED to said intervention.

Table III. Comparison of children with and without AKI

Variables	AKI (n = 83)	No AKI (n = 119)	P value
Age, y	8.1 ± 5.8	7.6 ± 5.5	.75
Sex, male, n (%)	45 (55)	60 (51)	.38
PRISM	10.7 ± 20.5	1.6 ± 2.0	<.001
PELOD	12.5 ± 8.5	6.6 ± 6.7	<.01
Previously healthy, n (%)	35 (42%)	48 (40%)	.88
Fluid given, mL/kg	47.2 ± 23.5	56.1 ± 17.3	.002
LOS, hospital, d	13.1 ± 12.8	8.9 ± 13.1	.013
LOS, PICU, d	5.2 ± 8.2	1.7 ± 1.9	<.001
RRT, n (%)	4 (5)	0	.016
30-d mortality, n (%)	11 (8.3)	2 (1.7)	.001

LOS when we controlled for AKI and PELOD (Table V; available at www.jpeds.com).

Discussion

A protocol-driven implementation of a SSP resuscitation bundle in the ED was associated with a 46% relative risk reduction of AKI development in children with clinical septic shock. The reduction in AKI was associated with decreased mortality, LOS, and need for RRT. There are at least 2 possible explanations for this: more rapid recognition of shock and more timely reversal of shock.

The time-sensitive nature of shock resuscitation is evident, with studies indicating greater mortality rates in children who have delay in reversal of shock.¹⁸⁻²⁰ One study demonstrated that early reversal of shock by community physicians was associated with improved outcomes.²¹ Similarly, children with septic shock resuscitated early (within 1 hour of arrival at ED) have decreased mortality²²; however, the ability to implement nationally endorsed guidelines often has been limited by the ability to recognize children while they are in compensated shock.²³⁻²⁵ In the absence of a protocol to avoid delays in patient recognition and timely intervention, adherence to published guidelines is suboptimal. Our shock protocol relied on a best practice alert, as previously reported, that reported temperature-corrected heart rate in addition to clinical diagnosis of septic shock alerting the clinicians to the presence of concerning vital sign changes at triage thereby facilitating early recognition.²⁶

Table IV. Comparison of patient outcomes before and after institution of the SSP

Variables	PRE SSP	POST SSP	P value
AKI	53 (54%)	30 (29%)	<.001
Risk	32	20	
Injury	16	7	
Failure	5	3	
Severe AKI	21 (21%)	10 (9.6%)	.03
RRT	4 (4%)	0	.037
LOS, hospital, d	15.3 ± 16.9	6.3 ± 5.1	<.001
LOS, PICU, d	4.5 ± 7.6	1.9 ± 2.3	<.001
Mortality, n (%)	10 (10)	3 (3)	.037

Pediatric septic AKI is a good illustration of the need for early identification and intervention, because a significant proportion of patients already have AKI on admission to the PICU, and earlier identification may have been the first reason (early recognition resulted in earlier treatment) that the AKI incidence declined after implementation of the SSP. Currently, no tool exists to identify patients at greatest risk of septic AKI to facilitate timely intervention. Serum creatinine is a late functional marker of AKI and is impacted by nutritional state, muscle mass, hydration, and—perhaps most importantly for pediatrics—age, making a single cutoff value inapplicable across age ranges. Electronic AKI alert tools relying on changes in serum creatinine have been created; however, these only detect AKI after it is already established. Moreover, mere identification of patients does not translate to improved outcomes in the absence of a guideline or protocol.²⁷ We have demonstrated that protocol-driven application of a resuscitation bundle in pediatric septic shock is associated with decreased renal injury. It seems that in pediatric septic AKI, the optimal intervention time is in the ED, even before a serum creatinine measurement is available.

The second reason for decreased AKI incidence was likely more timely fluid resuscitation. Although 3 recent large, randomized adult trials have raised questions about the specific components of early goal-directed therapy,²⁸⁻³⁰ there is no doubt that earlier reversal of shock is associated with improved outcomes both in children and adults. Although the total volume of normal saline administered as boluses was not substantially different between the PRE- and POST-SSP, what did differ was how quickly fluids were administered. The median time to intervention was 34 minutes in our shock protocol, almost one-half the time to first intervention in the control group.^{29,31} As part of the shock protocol preparatory education, the importance of clinical re-evaluation to assess for rales, hepatomegaly, and other clinical findings of fluid overload was emphasized to clinicians. Despite the very heterogeneous group of providers practicing in the ED of our quaternary care children's hospital, as well as the heterogeneous nature of our patient population, there was no evidence that more rapid delivery of fluid was associated with adverse outcomes. Instead, it is likely that more rapid recognition of shock by integration of a best practice alert, followed by protocolized fluid resuscitation, may have mitigated the development of AKI in these children. It is possible that the mortality rate and decreased LOS also was due to more rapid initiation of antimicrobial therapy. In one large PICU-based study, the odds of mortality increased almost 5-fold for delays in antibiotic receipt of more than 3 hours and children who received antibiotics promptly had more organ failure-free days.³²

There are several limitations. The intervention was not instituted in a randomized controlled fashion. Instead, patients from the same institution immediately preceding the implementation of the protocol were used as controls. Although there were no differences in severity of illness or burden of chronic comorbidities by the available scores tracked by our institution, it is possible that we have not

considered all potential confounders. As the study used a before/after design, we can infer associations, but not causation. Because the cohorts are from a consecutive period of 18 months, significant practice changes are unlikely. However, the management of patients was not standardized outside of the confines of the protocol. Use of the protocol was not mandatory and there were known protocol deviations with eligible patients not being enrolled; however, there was almost 90% adherence with the SSP during the study period.¹⁶ Not all variables were available for all children. These included time to renal recovery in AKI, urine output and fluid balance, and renal function at the time of hospital or PICU discharge. The fluid balance in the PICU was not assessed. Patients with CKD or ESRD were excluded from the study and the findings cannot be extrapolated to patients with preexisting renal dysfunction.

The incidence of septic AKI seen in our cohort was consistent with existing literature in pediatric severe sepsis.^{15,33} Septic AKI, the leading cause of AKI in the intensive care unit, has no effective treatment and dialysis fails to improve outcomes.^{17,34} Given the limited benefit of existing treatment modalities once AKI is established, one option to minimize the AKI burden would be to shift from a therapeutic to preventative stance. ■

Submitted for publication Apr 7, 2015; last revision received Jul 27, 2015; accepted Aug 21, 2015.

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Table V. Multivariate analyses of risk factors for the development of AKI

Variables	aOR (95% CI)	P value
Shock protocol	0.27 (0.13-0.56)	<.001
PELOD	1.08 (1.03-1.12)	.002
Age	1.01 (0.96-1.07)	.47
Sex	0.99 (0.54-1.84)	.99